

ATOMS AND MOLECULES

Class 9 | Science | Chapter 3

LAWS OF CHEMICAL COMBINATION

Law of Conservation of Mass

Mass can neither be created nor destroyed in a chemical reaction. Total mass of reactants = Total mass of products.

Law of Constant/Definite Proportions

In a chemical compound, the elements are always present in a fixed ratio by mass, no matter the source.

Ex: Water always has H:O mass ratio = 1:8.

ATOM

An atom is the smallest particle of an element that can take part in a chemical reaction; it may or may not exist independently.

- Atoms are extremely small -- sizes measured in nanometres ($1 \text{ nm} = 10^{-9} \text{ m}$).
- Symbols represent atoms of elements, e.g. H (Hydrogen), O (Oxygen), Na (Sodium), Fe (Iron).

ATOMIC MASS

Atomic mass is expressed in atomic mass units (u); $1 \text{ u} = 1/12$ th the mass of a Carbon-12 atom.

Ex: Atomic mass of Hydrogen = 1 u , Oxygen = 16 u , Carbon = 12 u .

MOLECULES & IONS

Formation, Types & Valency

MOLECULE

A molecule is the smallest particle of an element or compound capable of independent existence and showing all properties of that substance.

DIAGRAM: MOLECULES

H₂ (Element molecule)



O₂ (Element molecule)



H₂O (Compound molecule)



Atomicity

Number of atoms in a molecule -- e.g. Monatomic (He), Diatomic (O₂, H₂), Triatomic (O₃), Polyatomic (P₄).

IONS

An ion is a charged particle -- formed when an atom or group of atoms gains or loses electrons.

Type	Charge	Example
Cation	Positive (lost electrons)	Na ⁺ , Ca ²⁺ , Al ³⁺
Anion	Negative (gained electrons)	Cl ⁻ , O ²⁻ , SO ₄ ²⁻

VALENCY

Valency is the combining capacity of an element -- decides how many bonds/atoms it combines with.

Element	Valency
Hydrogen, Sodium, Chlorine	1
Oxygen, Calcium, Magnesium	2
Nitrogen, Aluminium	3
Carbon	4

CHEMICAL FORMULAE

Writing Formulae & Molecular Mass

WRITING A CHEMICAL FORMULA

- Write symbols of the elements/radicals side by side.
- Cross over the valencies (write the valency of one as subscript of the other).
- Simplify the ratio to lowest whole numbers where possible.

Compound	Formula
Water	H ₂ O
Ammonia	NH ₃
Carbon dioxide	CO ₂
Sodium chloride	NaCl
Calcium oxide	CaO
Aluminium oxide	Al ₂ O ₃

COMMON POLYATOMIC IONS

Ion	Formula	Charge
Hydroxide	OH ⁻	-1
Nitrate	NO ₃ ⁻	-1
Carbonate	CO ₃ ²⁻	-2
Sulphate	SO ₄ ²⁻	-2
Ammonium	NH ₄ ⁺	+1
Phosphate	PO ₄ ³⁻	-3

MOLECULAR MASS & MOLE CONCEPT

Molecular mass = sum of atomic masses of all atoms in a molecule.

Ex: Molecular mass of H₂O = 2(1) + 16 = 18u.

A mole is the amount of substance containing 6.022×10^{23} particles (Avogadro's number, N_A).

1 mole of any substance has a mass equal to its molar mass (in grams) -- e.g. 1 mole of water = 18 g.

SUMMARY & REVISION

Key Formulae + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Atom** -> smallest particle of an element
- **Molecule** -> group of atoms bonded together, independently stable
- **Ion** -> charged particle (atom/group that gained or lost electrons)
- **Valency** -> combining capacity of an element
- **Mole** -> 6.022×10^{23} particles of a substance
- **Molar mass** -> mass of 1 mole of substance in grams

KEY FORMULA BOX

Number of moles = Given mass / Molar mass

Number of particles = Moles $\times$ 6.022×10^{23}

SOLVED EXAMPLE

Q: Calculate the molecular mass of CO₂. (Atomic mass: C=12, O=16)

A: Molecular mass = $12 + 2(16) = 12 + 32 = 44$ u.

ONE-LINE RECAP FOR REVISION

Mass is conserved; compounds form in fixed ratios.

Atoms combine using valency to form molecules and compounds.

1 mole = 6.022×10^{23} particles = molar mass in grams.

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